

Movie Rating Prediction using Machine Learning

Project Objective

- Develop a machine learning model to predict movie ratings.
- Analyze movie-related features influencing ratings.
- Understand the complete ML workflow.
- Improve skills in preprocessing and predictive analytics.
- Learn regression on real-world datasets.

Dataset

- Source: Kaggle
- Format: CSV
- Features include Movie Name, Genre, Director, Actors, Duration, Language, Votes, Budget, Rating.

Programming Language

- Python

Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Project Workflow

- Import libraries.
- Load dataset.
- Inspect structure.
- Check missing values.
- Remove duplicates.
- Clean and preprocess data.
- Encode categorical variables.
- Perform EDA.
- Create charts and graphs.
- Select features.
- Split train/test data.
- Train regression model.
- Predict ratings.
- Evaluate using MAE, MSE, RMSE and R^2 .

Data Cleaning

- Removed duplicates.
- Handled missing values.
- Converted categorical data.
- Verified consistency.

Visualizations

- Histograms
- Bar Charts
- Scatter Plots
- Heatmaps
- Box Plots

Skills Learned

- Data Cleaning
- EDA

- Feature Engineering
- Regression
- Model Evaluation
- Python Programming

Applications

- Recommendation Systems
- Streaming Platforms
- Audience Analytics
- Entertainment Industry Decision Support

Future Enhancements

- Larger datasets
- Advanced algorithms
- Deep Learning
- Web deployment